

Manohar Mahadev Athani

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Professional Summary

Recently graduated ECE engineer from BIT Bangalore with strong foundations in DSA, backend development, Low-Level Design, and High-Level Design. Built production-grade Node.js microservices deployed on Google Cloud Platform (GCP), applied SOLID principles across real-world LLD projects, and designed scalable distributed architectures covering load balancing, caching, asynchronous processing, and cloud-native deployments.

Education

- Bangalore Institute of Technology** Bangalore, India
B.E. in Electronics and Communication | CGPA: 8.83 May 2026
- SRA Composite PU College** Banahatti, India
PUC | Percentage: 84% 2022

Experience

- Mindmatrix** Remote
Android App Development Intern — Generative AI 2026
- Built Android application features using **Kotlin** and **Jetpack Compose**, implementing clean UI components with **MVVM architecture** for clear separation of concerns.
 - Integrated **REST APIs** for real-time data fetching and managed local data persistence using **Room Database**, ensuring offline-first functionality.
 - Implemented dynamic list rendering with **RecyclerView** and optimized app performance through efficient state management in the MVVM pattern.

Projects

- Notification Service** | *Node.js, Express.js, BullMQ, Redis, PostgreSQL, JWT, Docker, GCP* [GitHub](#) | [Live Docs](#)
- Built a production-grade notification microservice supporting **Email, SMS, and In-App** delivery through a unified **REST API**, using **BullMQ** and **Redis** for asynchronous job processing with retry mechanisms.
 - Deployed on **Google Cloud Run** with **Cloud SQL (PostgreSQL)** and **Docker**, implementing **JWT authentication** and auto-scaling cloud infrastructure for reliable, high-availability message delivery.
- API Rate Limiter with Analytics** | *Node.js, Express.js, Redis, Lua, Docker* [GitHub](#)
- Built a distributed API rate limiting middleware implementing **Token Bucket**, **Sliding Window**, and **Fixed Window** algorithms using **Redis** and atomic **Lua scripts** for race-condition-free request handling.
 - Developed a real-time analytics dashboard tracking request volume, block rate, hourly traffic, and top blocked IPs; containerized with **Docker** and implemented a **fail-open** design for high availability.
- Distributed Task Scheduler** | *Node.js, Express.js, C++17, PostgreSQL, Google Pub/Sub, Cloud Run, Docker* [GitHub](#)
- Engineered a distributed task scheduler with a C++17 **min-heap** scheduler **O(1)** peek, **O(log n)** insert/delete integrated with **Node.js** services using **Google Pub/Sub** for reliable task execution.
 - Implemented a cloud-native deployment pipeline using **GitHub Actions**, **Artifact Registry**, and **Cloud Run** with **Cloud SQL**, **JWT**, and **Secret Manager** for secure, fault-tolerant execution.

Technical Skills

- Languages:** C, C++, JavaScript
- Backend & Web:** Node.js, Express.js, REST APIs, Microservices, Asynchronous Processing, WebSockets, HTML, CSS
- Databases & Caching:** PostgreSQL, MongoDB, Redis, SQL
- Message Queues:** BullMQ, Redis Queues, Event-Driven Architecture
- Cloud & DevOps:** Google Cloud Platform (GCP), Cloud Run, Cloud SQL, Docker, Containerization, CI/CD
- System Design:** Class Design, UML Diagrams, Design Patterns (Singleton, Factory, Strategy), Microservices, Load Balancing, Caching, DB Sharding, API Gateway, CAP Theorem, CDN
- Core Concepts:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks
- Principles & Tools:** SOLID, DRY, KISS, Git, Docker, VS Code, Linux

Extracurricular Activities

- Active member of the **Robotics Club**, Bangalore Institute of Technology.
- Participated in **Inter-college Coding Hackathon (2023)**.